Magnetic Vortex Nanodiscs Enable Remote Magnetomechanical Neural Stimulation

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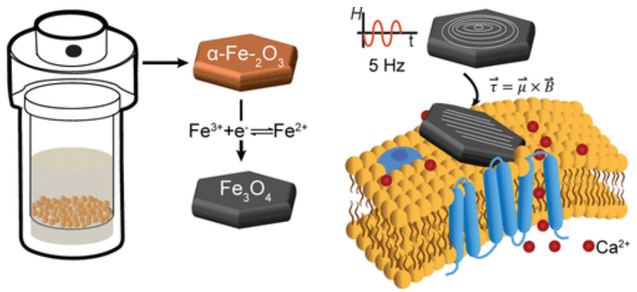
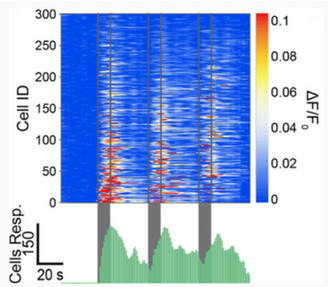


Introduction

- Magnetic nanoparticles (MNPs) have been used as remote transducers, due to the ability to manipulate them under an external MF.
- In this study, magnetite magnetic nanodiscs' (MNDs) transition from vortex to in-plane magnetization under an external MF was used to apply tension on dorsal root ganglion (DRG) neurons.
- Magnetite MNDs in different sizes were synthesized, characterized and coated with PMAO
- The MNDs were placed on DRG clotures and a moderate external MF was applied.
- Ca^{+2} levels were recorded to test the effect of MNDs torque on the cells.



Objective

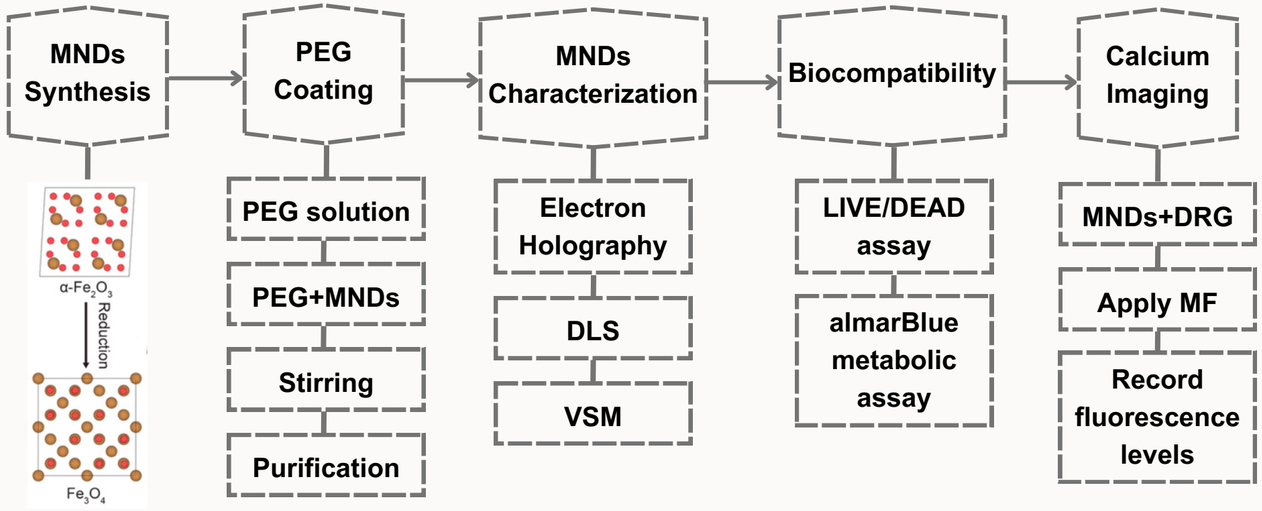
To explore a new coating that will improve:

- MNDs bonding to cells
- MNDs torque

And maintain:

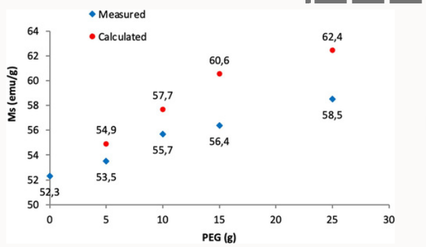
- MNDs magnetization
- Biocompatibility

Methods



Results

MNDs Magnetization & Torque



$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

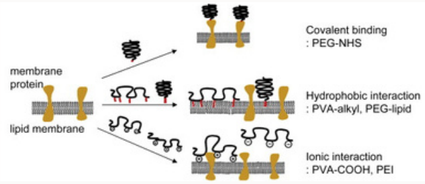
$$|\vec{\tau}| = |\vec{\mu}||\vec{B}| sin\theta_{\vec{\mu}-\vec{B}}$$

$$|\vec{\tau}|_{max} = |\vec{\mu}||\vec{B}|$$

$$|\vec{\mu}| = V \cdot \rho \cdot M_s$$

$$M_{s_{PEG}} = k \cdot M_s$$

$$|\vec{\tau}|_{PEG} = k \cdot |\vec{\tau}|$$

Biocompatibility

Bonding to Cells